INCH-POUND

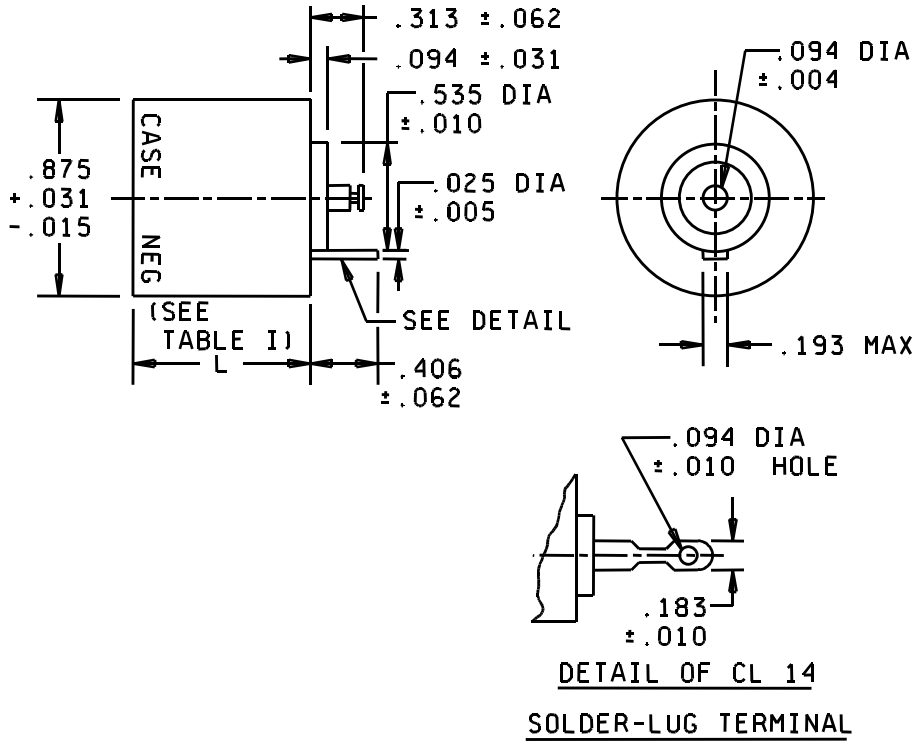
MIL-DTL-3965/1J
19 August 2002
SUPERSEDING
MIL-C-3965/1H
26 August 1970

DETAIL SPECIFICATION SHEET
CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM,
(POLARIZED, SINTERED SLUG) - 85°C (VOLTAGE DERATED TO 125°C),
STYLES CL14 AND CL16

INACTIVE FOR NEW EQUIPMENT DESIGN
NO SUPERSEDING DOCUMENT

This specification is approved for use to the Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-3965.

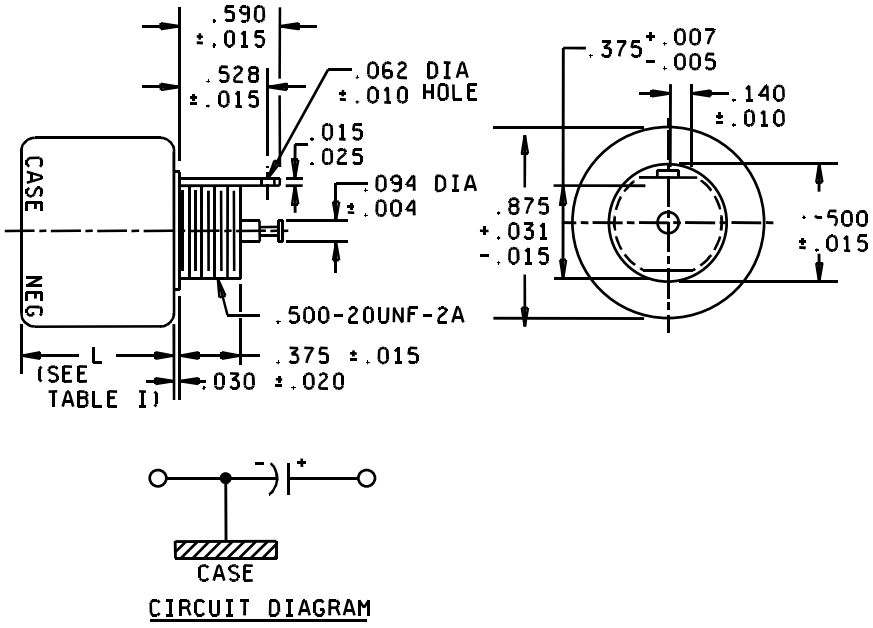


STYLE CL14

FIGURE 1. Styles CL14 and CL16 capacitors.

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STYLE CL16

Inches	mm	Inches	mm
.004	0.10	.140	3.56
.005	0.13	.183	4.65
.007	0.18	.193	4.90
.010	0.25	.313	7.95
.015	0.38	.375	9.53
.020	0.51	.406	10.31
.025	0.64	.500	12.70
.030	0.76	.528	13.41
.031	0.79	.535	13.59
.062	1.57	.590	14.99
.094	2.39	.875	22.23

NOTES:

1. All dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Styles CL14 and CL16 capacitors – Continued.

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TABLE I. Capacitor characteristics.

PIN 1/	Voltage			Capacitance	DC leakage (max)			Dissipation factor (max)	Impedance (max)	Dimension L ±.062 (1.57)
	Rated (+85°C)	Derated (+125°C)	Surge (+85°C)		+25°C	+85°C	+125°C			
	volts, dc	volts, dc	volts,dc	µF	µA	µA	µA	%	ohms	Inches
CL1-BF101UPG	20	18	23.0	100.0	10	50	80	21.0	30	.540 (13.72)
CL1-BF201UPG	20	18	23.0	200.0	16	80	128	36.0	20	.732 (18.59)
CL1-BH750UPG	30	26	34.5	75.0	11	55	88	15.0	45	.540 (13.72)
CL1-BH151UPG	30	26	34.5	150.0	13	90	104	29.0	30	.732 (18.59)
CL1-BK400UPG	60	52	69.0	40.0	12	60	96	8.2	65	.540 (13.72)
CL1-BK800UPG	60	52	69.0	80.0	19	95	152	16.0	35	.732 (18.59)
CL1-BM250UPG	90	78	103.0	25.0	11	55	88	5.1	90	.540 (13.72)
CL1-BM500UPG	90	78	103.0	50.0	18	90	144	10.0	45	.732 (18.59)
CL1-BR120UPG	180	155	207.0	12.0	11	55	88	5.1	180	.920 (23.37)
CL1-BR250UPG	180	155	207.0	25.0	18	90	144	10.0	90	1.300 (33.02)
CL1-BS080UPG	270	235	310.0	8.0	11	55	88	5.1	270	1.270 (32.36)
CL1-BS160UPG	270	235	310.0	16.0	18	90	144	10.0	135	1.865 (47.37)
CL1-BT060UPG	360	310	414.0	6.0	11	55	88	5.0	360	1.635 (41.53)
CL1-BT120UPG	360	310	414.0	12.0	18	90	144	10.0	180	2.420 (61.47)
CL1-BU050UPG	450	390	518.0	5.0	11	55	88	4.9	450	2.000 (50.80)
CL1-BU100UPG	450	390	518.0	10.0	18	90	144	9.8	225	2.980 (75.69)
CL1-BV040UPG	540	470	621.0	4.0	11	55	88	5.1	540	2.365 (60.07)
CL1-BV080UPG	540	470	621.0	8.0	18	90	144	10.0	270	3.532 (89.71)
CL1-BW3R5UPG	630	545	724.0	3.5	11	55	88	5.0	630	2.720 (69.09)
CL1-BW070UPG	630	545	724.0	7.0	18	90	144	10.0	315	4.062(103.17)

1/ Complete type designation will include an additional symbol to indicate style.

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TABLE II. Cross-reference data.

From	To
PIN 1/ MIL-DTL-3965/1	PIN 2/ MIL-PRF-39006/19
	M39006/19-
CL1-BF101UPG	1000
CL1-BF201UPG	1001
CL1-BH750UPG	1002
CL1-BH151UPG	1003
CL1-BK400UPG	1004
CL1-BK800UPG	1005
CL1-BM250UPG	1006
CL1-BM500UPG	1007
CL1-BR120UPG	1008
CL1-BR250UPG	1009
CL1-BS080UPG	1010
CL1-BS160UPG	1011
CL1-BT060UPG	1012
CL1-BT120UPG	1013
CL1-BU050UPG	1014
CL1-BU100UPG	1015
CL1-BV040UPG	1016
CL1-BV080UPG	1017
CL1-BW3R5UPG	1018
CL1-BW070UPG	1019

1/ Complete type designation will include an additional symbol to indicate style.
2/ PIN referenced will substitute for styles CL14 and CL16.

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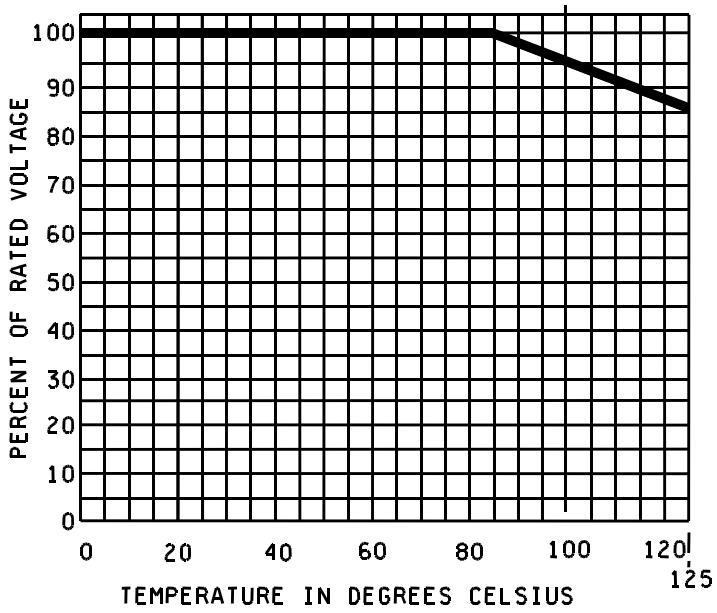


FIGURE 2. Voltage derating with temperature.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Tubular; uninsulated.

Seal type: Hermetic (symbol G).

Terminals: Solder-lug.

Rated temperature: -55°C to +85°C, voltage derated to +125°C (see figure 2).

Rated voltage: See table I.

Capacitance (cap) (nom): See table I.

Capacitance tolerance: Symbol U (+75 percent, -15 percent).

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DC leakage (DCL) (max) (at +25°C): See table I.

Dissipation factor (DF) (max): See table I.

Seal: In accordance with MIL-DTL-3965.

Dielectric withstanding voltage (DWV):

Sleeving to case (min): Not applicable.

Terminal to case: Not applicable.

Insulation resistance:

Sleeving to case (min): Not applicable.

Terminal to case: Not applicable.

Temperature cycling:

DCL (max): 125 percent of 25°C value of table I.

ΔCap.: Within $\pm$ 5 percent of initial measurement.

DF (max): 115 percent of table I value.

Solderability: Not applicable.

Terminal strength:

Pull test: In accordance with MIL-DTL-3965.

Bend test (if required): In accordance with MIL-DTL-3965.

Torque test (CL16 only): In accordance with MIL-DTL-3965.

Wire-lead bend test: Not applicable.

Twist test: Not applicable.

Surge voltage: See table I.

Moisture resistance:

DCL (max): 125 percent of 25°C value of table I.

ΔCap.: Within $\pm$ 8 percent of initial measurement.

DF (max): 115 percent of table I value.

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Low temperature (storage):

DCL (max): See table I (25°C).

ΔCap.: Within ± 5 percent of initial measurement.

DF (max): See table I.

Stability at low and high temperature:

Step 1 (+25°C):

DCL (max): See table I.

Cap.: Within tolerance of table I value.

DF (max): See table I.

Step 2 (-55°C):

Impedance: See table I.

ΔCap. (max): -60 percent (20 V), -45 percent (30 V), or –35 percent (>30 V) of step 1 value.

Step 3 (+25°C):

DCL (max): See table I.

ΔCap.: Within ±5 percent of step 1 value.

DF (max): See table I.

Step 4 (+85°C):

DCL (max): See table I.

ΔCap (max): +15 percent (≥ 30 V) or +10 percent (> 30 V) of step 1 value.

DF (max): See table I.

Step 5 (+125°C):

DCL (max): See table I.

ΔCap. (max): +20 percent (≥ 26 V) or +10 percent (>26 V) of step 1 value.

DF (max): See table I.

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Step 6 (+25°C):

DCL (max): See table I.

ΔCap.: Within ±5 percent of initial measurement.

DF (max): See table I.

Reverse voltage: Not applicable.

Life at +85°C:

2,000 hours (qualification):

DCL (max): See table I (25°C).

ΔCap.: Within ±15 percent of initial measurement.

DF (max): See table I.

240 hr (group B):

DCL (max): See table I (25°C).

ΔCap.: Within ±10 percent of initial measurement.

DF (max): See table I.

1,760 hr (group C):

DCL (max): See table I (25°C).

ΔCap.: Within ±15 percent of initial measurement.

DF (max): See table I.

Barometric pressure:

Test voltage: Rated voltage for 20 V to 450 V units; 500 V dc for 540 V and 630 V units.

Life at +125°C.

DCL (max): See table I (25°C).

ΔCap.: Within ±15 percent of initial measurement.

DF (max): 130 percent of initial requirement.

DWV and IR: Not applicable.

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Marking: Capacitors shall be marked as shown in the following example:

EXAMPLE:

CL14BF101UPG	-	Type designation
12345 7015	-	Source and date code
100 μF 20 V dc	-	Capacitance and rated voltage (at 85°C).

Polarity: Capacitors shall be marked “CASE NEG” on the negative end (see figure 1).

Voltage groups (85°C rated voltage):

I - - - - - - - - - -	20 V to 90 V inclusive
II - - - - - - - - - - -	180 to 360 V, inclusive
III- - - - - - - - - -	450 V to 630 V, inclusive.

NOTE: The capacitors covered by this specification sheet are substitutable for those formerly covered by MIL-C-3965/1. See table II.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA – CC

Preparing activity:
DLA - CC

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